Computational workflow

Shared field conventions; separate state equations

1 Set parameters

Pulse parameters and units
Momentum grid p/m
Tail extent L/τ
Solver tolerances

2 Define the field

Sauter pulse or pulse train
E(t), A(t), finite window

3 Solve each mode

Dirac spinor: DOP853
QKE state (f,u,v): DOP853
Separate integrations

4 Validate each run

Exact Sauter benchmark
Final norm and invariant
Dirac–QKE agreement
Tail-window scan

5 Save datasets

Spectrum and yield CSV
Tail and p–L/τ map CSV
Run metadata JSON

6 Plot and verify

Python/Matplotlib figures
PDF and PNG outputs
Package checksums

reproduce.py · make_reproducibility_figures.py

redraw_figures.py · verify_checksums.py